DETAILED WEBSITE ANALYSIS REPORT

Introduction

Web technologies have transformed the way users learn, shop, access government services, and prepare for careers. Modern websites are designed using advanced front-end frameworks, secure back-end systems, and cloud infrastructures to ensure performance, scalability, and security. This report presents a comprehensive analysis of five widely used websites: TryHackMe, Election Commission of India (ECI), Myntra, GeeksforGeeks, and LeetCode.

Each website is examined based on the following parameters:

- URL
- Purpose
- Domain
- Inventor/Founder
- Technology used
- Features
- Design and usability
- Advantages
- Limitations
- Detailed website summary

The objective is to understand how these platforms function, what technologies they use, and how effectively they serve users.

Website 1 – TryHackMe

URL

https://tryhackme.com

Purpose

TryHackMe is an online cybersecurity learning platform that provides hands-on training in ethical hacking, penetration testing, and network security.

Domain

Education / Cybersecurity

Inventor / Founder

Founded by Ben Spragg in 2018.

Technology Used

- React.js (Frontend)
- Node.js (Backend)
- Linux-based virtual machines
- Cloud hosting (AWS/Azure)
- APIs
- Docker containers
- Database systems for user tracking
- HTTPS security protocols

Features

- Interactive hacking labs
- Capture The Flag (CTF) challenges
- Learning paths for beginners to advanced users
- Virtual machines for real-time practice
- Certificates and progress tracking
- Community discussions

Design and Usability

The interface is modern and visually attractive. The dashboard clearly displays rooms, challenges, and progress. Color coding helps users identify difficulty levels. Navigation is mostly intuitive, but beginners may feel overwhelmed due to many options.

Advantages

- Practical learning environment
- Beginner-friendly tutorials
- Real-world attack simulations
- Affordable compared to other platforms
- Strong community support

Limitations

- Some labs require premium subscription
- VM startup may be slow
- Heavy resource usage on low-end systems

Website Summary

TryHackMe plays an important role in cybersecurity education by bridging the gap between theory and practice. Instead of only reading about attacks, users can perform real hacking tasks inside safe environments. This experiential learning approach improves technical skills and problem-solving abilities. The platform continues to grow with new content, making it highly valuable for students and security professionals.

Website 2 – Election Commission of India (ECI)

URL

https://eci.gov.in

Purpose

Provides official election information, voter services, and public announcements.

Domain

Government / Public Service

Inventor / Founder

Developed and maintained by the Election Commission of India (Government body).

Technology Used

- HTML, CSS, JavaScript
- Government web frameworks
- Secure servers
- Database systems
- Form handling systems
- HTTPS encryption

Features

- Voter registration services
- Election schedules
- Results publication
- Candidate information
- Public notices
- Downloadable forms

Design and Usability

The website is functional but has an outdated design. It focuses more on information rather than aesthetics. The layout contains large text blocks and limited visual hierarchy. Navigation can be confusing.

Advantages

- Authentic and official data
- Secure and reliable
- Important public services
- Accessible to citizens nationwide

Limitations

- Old interface design
- Slow navigation
- Not very mobile-friendly
- Heavy text content

Website Summary

The ECI website serves as a critical platform for democratic processes in India. Its primary goal is to deliver trustworthy information rather than visual appeal. While the design may appear outdated, the reliability and authenticity make it highly valuable. Improving UI/UX could significantly enhance user experience.

Website 3 – Myntra

URL

https://www.myntra.com

Purpose

Online shopping platform for fashion and lifestyle products.

Domain

E-commerce / Retail

Inventor / Founder

Founded by Mukesh Bansal, Ashutosh Lawania, and Vineet Saxena (2007).

Technology Used

- React/Angular frontend

- Java/Spring backend
- Microservices architecture
- Cloud hosting
- Payment gateways
- Recommendation algorithms
- Databases

Features

- Product catalog
- Search and filters
- Secure checkout
- Offers and discounts
- Order tracking
- Easy returns

Design and Usability

The interface is colorful and attractive. Product images are clear and filters help users find items quickly. However, too many advertisements and banners may distract users.

Advantages

- Large variety of products
- Smooth shopping experience
- Multiple payment options
- Easy return policy

Limitations

- Pop-ups and ads
- Heavy pages may load slowly
- Occasional price fluctuations

Website Summary

Myntra successfully combines fashion retail with digital convenience. Advanced recommendation systems personalize the shopping experience. Despite some design clutter, it remains one of India's most popular fashion e-commerce platforms.

Website 4 – GeeksforGeeks

URL

https://www.geeksforgeeks.org

Purpose

Provides programming tutorials and interview preparation resources.

Domain

Education / Programming

Inventor / Founder

Founded by Sandeep Jain (2008).

Technology Used

- HTML, CSS, JavaScript
- Backend server technologies
- Databases
- CMS systems
- Online compilers

Features

- Tutorials
- Coding problems
- Practice questions
- Interview guides
- Courses
- Articles

Design and Usability

Content is well structured but pages are crowded with ads. Sidebars and banners reduce readability. The site is informative but visually cluttered.

Advantages

- Huge content library
- Free tutorials
- Beginner-friendly explanations

- Good for interview preparation

Limitations

- Too many advertisements
- Cluttered layout
- Inconsistent formatting

Website Summary

GeeksforGeeks is one of the most trusted platforms for programming learners. Its strength lies in detailed explanations and numerous practice problems. Reducing ads and improving design would greatly enhance user satisfaction.

Website 5 – LeetCode

URL

https://leetcode.com

Purpose

Coding practice and technical interview preparation.

Domain

Education / Competitive Programming

Inventor / Founder

Founded by Winston Tang (2015).

Technology Used

- React frontend
- Backend services
- Online compiler engine
- Cloud servers
- Databases
- Contest systems

Features

- Algorithm problems
- Weekly contests
- Company-specific questions
- Online compiler
- Discussion forums

- Rankings

Design and Usability

Clean and professional interface. Problem statements are clear. Filters help users sort questions. Some features are locked for premium users.

Advantages

- High-quality problems
- Real interview preparation
- Competitive environment
- Strong community

Limitations

- Many paid features
- Difficult for beginners
- Limited explanations in some problems

Website Summary

LeetCode is highly effective for improving coding skills and preparing for technical interviews. Its competitive environment encourages consistent practice. Despite paywalls, it remains one of the best platforms for serious learners.

Overall Website Analysis Summary

SR NO	WEBSITE URL	PURPOSE OF WEBSITE	THINGS LIKED IN WEBSITE	THINGS DISLIKED IN WEBSITE	(GOOD / BAD)
1	https://tryhackme.com	Cybersecurity learning platform for ethical hacking and hands-on labs.	Interactive labs, beginner friendly, practical exercises, real-world scenarios.	Dashboard can feel cluttered for beginners; too many learning paths shown at once; VM launch sometimes slow; navigation between rooms/labs not always intuitive; mobile experience limited for lab usage.	GOOD
2	https://eci.gov.in	Official Election Commission of India site for election updates, results, and voter services.	Authentic information, voter services, downloadable forms, secure.	Outdated interface design; excessive text blocks; poor visual hierarchy; difficult navigation structure; limited search usability; not very mobile responsive; slow page loading.	BAD
3	https://www.myntra.com	Online fashion and lifestyle shopping platform.	Attractive UI, filters, discounts, easy returns, multiple brands.	Too many banners and pop-ups; distracting promotional ads; heavy graphics increase load time; filter panel sometimes complex; overcrowded product pages.	GOOD
4	https://www.geeksforgeeks.org	Educational platform for programming tutorials, coding practice, and interview prep.	Large content library, examples, coding problems, free resources.	Overloaded with ads and sidebars; cluttered layout; inconsistent font sizes; difficult readability; too many links on a single page; weak visual focus on main content.	GOOD

5.	https://leetcode.com	Coding practice platform for algorithm and data structure problems.	High-quality problems, contests, interview preparation, clean interface.	Some features hidden behind menus; beginners may find UI confusing; problem filtering not always clear; many premium locks reduce accessibility; discussion pages look dense and less organized.	GOOD